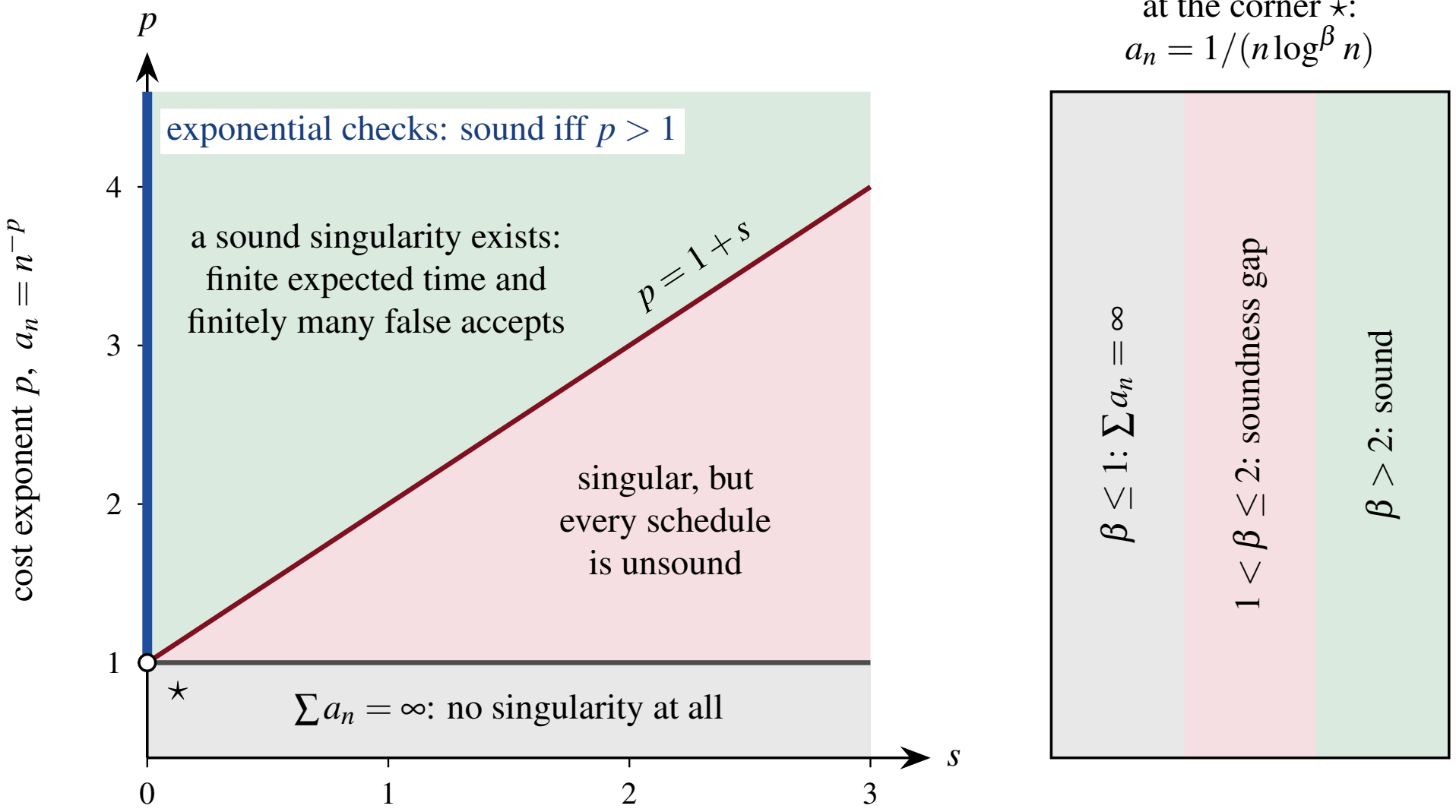

p
exponential checks: sound iff $p > 1$
a sound singularity exists:
finite expected time and
finitely many false accepts
$p = 1 + s$
singular, but
every schedule
is unsound
$\star$
$\sum a_n = \infty$: no singularity at all
cost exponent p, $a_n = n^{-p}$
4
3
2
1
0
1
2
3
s
order s of the check: a wrong step passes r runs with probability $\leq Cr^{-1/s}$
at the corner $\star$:
$a_n = 1/(n \log^{\beta} n)$
$\beta \leq 1$: $\sum a_n = \infty$
$1 < \beta \leq 2$: soundness gap
$\beta > 2$: sound